Assignment #3

Camera Calibration

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Submit By: 11:59pm September 20, 2024
revised August 26, 2024

1 Overview

For this assignment your goal is to find the calibration parameters for the camera you plan to use in this course. For this assignment you will need to be able to read and write to a JSON file, read/write image files, and use some functions from OpenCV.

In general, you should assume that any camera you are taking images from will need distortion correction. If you write your code well you may be able to reuse it in future assignments.

2 Instructions

1. Print out and mount some kind of calibration pattern. Two of the OpenCV provided patterns can be found here:
   - **Chessboard:** https://raw.githubusercontent.com/opencv/opencv/master/doc/pattern.png
   - **Circles:** https://raw.githubusercontent.com/opencv/opencv/master/doc/acircles_pattern.png
2. Using the target camera, take at least 10 photos of your pattern board in various angles. Note that each image must be one where OpenCV's `findChessboardCorners(...)` is able to detect features. You can verify that a feature was detected using `drawChessboardCorners(...)`.
   - Additional functionality you might want(not required):
   - *Automatic capture – automatically capture/use an image if chessboard pattern is recognized.*
   - *A counter telling you how many calibration images have been collected.*
   - *Calculate and display the reprojection error.*

3. Calibrate your camera and save the parameters to a JSON file `camera-params.json` or something similar.
    - **JSON Syntax:** https://www.json.org/json-en.html
    - **Python JSON Library:** https://docs.python.org/3/library/json.html
4. Use Python to load in the parameters from the JSON file.
5. Capture an image from the camera and save it as `<image-name>_distorted.png`.
6. Use the parameters loaded in step 3 to undistort the image captured in step 4. Save this image as `<image-name>_undistorted.png`.
7. Make a brief write-up(about one paragraph) of what your code does. If you have any notes or advice that you wish you could go back in time and give yourself before the assignment, add that too.

Note: *Don't put this off – this assignment can be more difficult than it looks. OpenCV's function parameters, object types, and return values are not always intuitive. You may want to save your work for this assignment and others into a library file for reuse in later assignments. I'm not too concerned if your code looks a lot like the tutorial code (this a pretty basic operation). But make sure you credit the tutorial code.*

3 Requirements

Make sure your zipped submission contains:

1. Source code `.py` file *with your name at the top.*
2. Your JSON file, `<camera-params>.json`.
3. Both `<filename>_distorted.png` and `<filename>_undistorted.png`.
4. Your write-up as a PDF.
    - ***Note:*** *If you are looking for an easy way to convert your write-up from something like Markdown to PDF, you might be interested in pandoc.*

4 Submission Instructions

Submit via Canvas or to my email (grwells@uidaho.edu). Most assignments require a write-up and source code, both submitted in a compressed/zipped folder containing all files required to run. Write-ups must be submitted in PDF format. When submitting an assignment via email, please put the course number and the assignment name in the subject line. Submission requirements may be amended in assignment instructions.